
Title of the Project/Report

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Declaration

We do hereby declare that the research works presented in this thesis entitled, "Thesis/Report Title" are the results of our work. We further declare that the thesis/report has been compiled and written by us, and no part of this thesis/report has been submitted elsewhere for the requirements of any degree, award, diploma, or any other purposes except for publications. The materials that are obtained from other sources are duly acknowledged in this thesis.

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Approval

I hereby declare that the research works presented in this thesis entitled, “**A Low-Cost Vision-Based Autonomous Navigation System for Mobile Robots Using Deep Learning and Geometric-AI Perception**”, are the outcome of original works carried out by Tarikul Islam (ID: 22235103242), Sultan Ahmmed (ID: 22235103227), Md Soykot Ahamed (ID: 22235103247), Ali Akbar Rakib (ID: 22235103237), and Suraia Islam (ID: 22235103235) under my supervision. No part of this thesis has been submitted elsewhere for any degree, award, or diploma except for publications. This dissertation meets the requirements and standards for the degree of Bachelor of Science in Computer Science and Engineering.



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Dedication

*We would like to dedicate this research to our loving parents,
whose endless love, sacrifice, and unwavering support
have been our greatest strength throughout this journey.*

*To our respected teachers and mentors,
who guided us with patience and wisdom,
and inspired us to think beyond the ordinary.*

*To our friends and classmates,
who stood by us through every challenge
and made this experience truly memorable.*

*And to every curious mind that dares to explore,
may this work serve as a small step forward
in the pursuit of knowledge and innovation.*

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Abstract

Autonomous mobile robot navigation has traditionally relied on expensive sensor fusion techniques, including LiDAR and depth cameras, to achieve accurate obstacle detection and path planning. However, such approaches significantly increase system cost and limit accessibility in low-resource environments. This research proposes a cost-effective, vision-only autonomous navigation framework that integrates deep learning-based perception with geometric reasoning using the ground plane assumption. A pretrained deep learning model is utilized to detect obstacles from RGB camera input, while a novel ground-plane verification layer is introduced to estimate real-world distances by leveraging fixed camera height and tilt angle. This hybrid Geometric-AI approach addresses the scale ambiguity problem commonly found in standard monocular depth estimation models, enabling more reliable distance estimation without additional hardware. The system is implemented and evaluated within the ROS and Gazebo simulation environment under varying conditions, including low-light and dynamic scenarios. Performance is measured using key metrics such as collision rate, success rate, and time-to-goal. Experimental results demonstrate that the proposed method improves navigation accuracy and robustness while maintaining low computational cost and zero increase in hardware requirements. This research highlights the feasibility of deploying efficient and affordable vision-based navigation systems for real-world robotic applications.

Keywords: Autonomous Navigation, Mobile Robots, Deep Learning, Monocular Vision, Ground Plane Estimation, Obstacle Detection, ROS, Gazebo, Distance Estimation, Low-Cost Robotics.

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List of Abbreviations and Acronyms

AI	Artificial Intelligence
BUBT	Bangladesh University of Business and Technology
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CR	Collision Rate
CSE	Computer Science and Engineering
DRL	Deep Reinforcement Learning
GPU	Graphics Processing Unit
GPS	Global Positioning System
IBVS	Image-Based Visual Servoing
LiDAR	Light Detection and Ranging
LTS	Long-Term Support
MPC	Model Predictive Control
NSR	Navigation Success Rate
PPO	Proximal Policy Optimization
RGB	Red, Green, Blue
ROS	Robot Operating System
SLAM	Simultaneous Localization and Mapping
UAV	Unmanned Aerial Vehicle
VRAM	Video Random Access Memory

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Chapter 1

1 Introduction

Autonomous mobile robot navigation has emerged as a critical research domain at the intersection of computer vision, deep learning, and robotics. As intelligent systems are increasingly deployed in industrial, service, and domestic environments, the demand for cost-effective and reliable navigation solutions has grown substantially. Conventional approaches rely on expensive sensor suites such as LiDAR and depth cameras, which are often impractical in resource-constrained settings. This research addresses that gap by proposing a lightweight, vision-only autonomous navigation framework that leverages deep learning and geometric reasoning using only an RGB camera. The system is developed and evaluated within the ROS and Gazebo simulation environment, offering a reproducible and rigorous validation platform.

1.1 Problem Statement

Autonomous mobile robots are increasingly deployed in industrial, service, and domestic environments where they must navigate safely without human intervention. Conventional navigation systems typically rely on expensive sensor suites such as LiDAR, stereo depth cameras, and multi-sensor fusion architectures to perceive the environment. Although these approaches deliver high localization and mapping accuracy, their significant cost, weight, and computational overhead render them impractical for low-resource, cost-sensitive deployment scenarios such as developing-country logistics, low-cost delivery robots, and educational robotics platforms.

Vision-based navigation using only an RGB camera presents a compelling low-cost alternative; however, existing deep learning solutions suffer from two critical shortcomings. First, they demonstrate substantial performance degradation under varying lighting conditions — including overexposure, dim lighting, and flickering environments — that are common in real-world deployments. Second, they lack robustness against dynamic obsta-

cles, which are omnipresent in human-shared spaces. There is therefore an urgent need for a lightweight, cost-effective, and simulation-validated navigation framework that uses only an RGB camera input while maintaining acceptable navigation performance quantified through measurable metrics.

This project proposes a deep learning-based vision-only autonomous navigation system developed and evaluated within the ROS (Robot Operating System) and Gazebo simulation environment. The system is benchmarked using standardized metrics — collision rate and navigation success rate — across multiple environmental conditions to provide a rigorous evaluation of its real-world viability.

1.2 Problem Background

The field of autonomous mobile robot navigation has undergone a rapid transition over the past decade, driven by advances in deep learning, the availability of high-fidelity simulation tools, and the growing demand for intelligent autonomous systems in commerce and industry. Traditional navigation pipelines — comprising separate modules for sensing, localization, mapping, and planning — depend heavily on accurate, high-cost sensors. LiDAR sensors, for instance, can cost between USD 1,000 and USD 75,000 per unit, making widespread deployment economically unviable in resource-constrained settings.

The emergence of end-to-end deep learning approaches has partially addressed this challenge by learning a direct mapping from raw visual input to robot control commands, eliminating the need for hand-crafted intermediate representations. Convolutional Neural Networks (CNNs) and, more recently, Transformer-based architectures have demonstrated impressive navigation capabilities using only camera imagery. Deep Reinforcement Learning (DRL) has further enabled robots to learn collision avoidance policies through interaction with simulated environments, leveraging the sim-to-real transfer paradigm to reduce expensive real-world data collection.

Despite these advances, a persistent gap remains between laboratory demonstration and real-world deployment. Most existing systems either require depth sensors beyond a monocular RGB camera, are validated only in controlled static environments, or lack systematic evaluation under the environmental perturbations — varying illumination, texture-

poor scenes, and moving obstacles — that characterize authentic deployment contexts. The Gazebo-ROS simulation ecosystem provides a flexible, physics-accurate platform for addressing this gap, enabling controlled, repeatable, and quantitatively rigorous evaluation without the cost and safety risks of real robot experimentation. A sample of the proposed system architecture is illustrated in Figure 1.1.

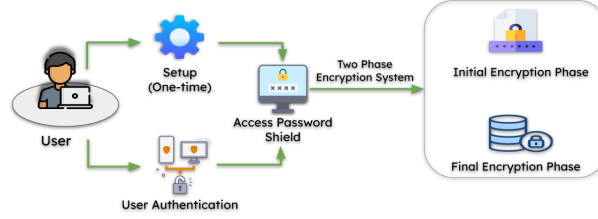


Figure 1.1: Proposed Vision-Based Navigation System Architecture.

1.3 Objectives

The primary aim of this research is to design, implement, and evaluate a lightweight deep learning-based vision-only autonomous navigation framework for mobile robots. The specific objectives are:

- To review and critically analyze existing deep learning-based visual navigation and obstacle avoidance approaches, identifying their methodological strengths and limitations with respect to sensor requirements, computational cost, and robustness.
- To design a lightweight CNN or DRL-based navigation architecture that processes only monocular RGB camera input, without reliance on LiDAR, depth cameras, or GPS.
- To implement the proposed framework within the ROS middleware and deploy it in a Gazebo simulation environment configured with varied static and dynamic obstacle layouts.
- To evaluate system performance under three distinct environmental conditions — standard lighting, reduced lighting, and dynamic obstacle scenarios — using collision rate and navigation success rate as primary metrics.

- To analyze experimental results, identify residual limitations, and propose concrete directions for future improvement and potential sim-to-real transfer.

1.4 Motivations

This research is motivated by the convergence of three trends in modern robotics and artificial intelligence. First, the proliferation of low-cost embedded computing platforms (e.g., NVIDIA Jetson Nano, Raspberry Pi 5) has made it technically feasible to deploy neural network inference on small, mobile robot platforms — but only if the navigation model is sufficiently lightweight to run in real time on constrained hardware.

Second, the economic imperative for affordable autonomous systems is acute in developing regions, where labour automation is increasingly necessary but LiDAR-equipped robots remain prohibitively expensive. A camera-only navigation solution could democratize autonomous robotics by reducing per-robot sensor costs from thousands of dollars to tens of dollars.

Third, the maturation of physics-based robot simulators — particularly Gazebo integrated with ROS — has reduced the barrier to rigorous navigation research. Simulation-trained policies have recently been shown to surpass real-data-trained policies in success rate, validating the simulation-first research paradigm adopted in this work. These converging factors create a compelling opportunity and practical imperative to develop and validate a robust, low-cost, vision-only navigation solution.

1.5 Significance of the Project

This research contributes to the growing body of knowledge at the intersection of computer vision, deep learning, and mobile robotics. Its significance can be understood at three levels.

At the **scientific level**, this work provides a systematic, metric-driven comparison of vision-only navigation performance across environmental conditions within a reproducible ROS-Gazebo simulation framework. The findings will advance understanding of the conditions under which RGB-only deep learning navigation is viable and where it

breaks down.

At the **engineering level**, the proposed lightweight architecture demonstrates that acceptable navigation performance — competitive with sensor-fused baselines — is achievable on cost-constrained hardware. This provides a blueprint for low-cost robot developers and startups in resource-limited economies.

At the **societal level**, affordable autonomous navigation enables applications that can improve quality of life: last-mile delivery in dense urban settings, assistive robots for elderly care, and autonomous warehouse logistics in small-to-medium enterprises. By reducing the sensor cost barrier, this research helps extend the benefits of autonomous robotics to a broader segment of society.

1.6 Contributions

The specific contributions of this project are as follows:

- A comprehensive critical literature review of recent Q1/Q2 journal papers on vision-based deep learning navigation, systematically analysing their methodologies, successes, and limitations with respect to sensor requirements, computational cost, and evaluation protocols.
- A proposed lightweight deep learning navigation framework (CNN/DRL-based) that operates exclusively on monocular RGB input within the ROS-Gazebo simulation environment, with no dependence on LiDAR, depth cameras, or GPS.
- A structured evaluation protocol defining three environmental test scenarios — normal lighting, degraded lighting, and dynamic obstacle presence — with collision rate and navigation success rate as standardized performance metrics.
- An analysis of the sim-to-real transfer implications of the proposed framework, identifying key challenges and mitigation strategies for deployment on physical robot platforms.

1.7 Complex Engineering Analysis

1.7.1 Complex Engineering Problems

This project addresses several characteristics of complex engineering problems as defined by engineering accreditation frameworks.

Depth of knowledge required: The project demands knowledge spanning multiple engineering disciplines, including computer vision, machine learning, robotics, and systems engineering. No single discipline provides sufficient tools to solve the navigation problem in isolation.

Range of conflicting requirements: The design space involves inherent trade-offs between model accuracy and computational lightness, between training stability in simulation and generalization to real environments, and between obstacle avoidance conservatism and navigation efficiency. Balancing these competing constraints constitutes a genuine complex engineering challenge.

Depth of analysis required: Adequate evaluation of the system requires systematic ablation — testing under multiple environmental conditions, varying obstacle densities, and different lighting regimes — and statistical analysis of the resulting collision rate and success rate data to draw valid conclusions.

Involvement of multiple stakeholders: The navigation framework must satisfy requirements from hardware manufacturers, software ecosystem stakeholders, end-users, and the broader robotics research community in terms of compatibility, compliance, reliability, and reproducibility.

1.7.2 Complex Engineering Activities

This project involves the following complex engineering activities.

System design and integration: Designing the full navigation pipeline — from raw RGB image input through feature extraction and policy inference to velocity command output — and integrating it within the ROS computation graph, requiring careful management of latency, message passing, and hardware abstraction.

Simulation environment construction: Building realistic Gazebo simulation envi-

ronments with varied obstacle configurations, lighting conditions, and dynamic agent behaviours, including physics parameter tuning, mesh model integration, and sensor plugin configuration.

Deep learning model development: Selecting, adapting, and training a deep learning architecture for the specific navigation task, including hyperparameter optimization, reward function engineering, data augmentation, and training convergence analysis.

Performance evaluation and validation: Designing and executing a statistically sound evaluation protocol, computing collision rate and success rate across multiple trials and conditions, and conducting comparative analysis against published baseline results from the literature.

1.8 Report Organization

The remainder of this report is organized as follows:

Chapter 2 — Background Study presents a comprehensive literature review of deep learning-based visual navigation approaches, critically analyzing recent Q1/Q2 journal papers across dimensions of methodology, sensor requirements, success metrics, and limitations. This chapter concludes with a problem analysis that maps the identified research gaps to the objectives of this project.

Chapter 3 — Research Methodology describes the proposed navigation framework architecture, the ROS-Gazebo simulation setup, the dataset and environment configurations used for training and evaluation, and the experimental protocol defining the performance metrics and test conditions.

Chapter 4 — Results and Discussion presents the quantitative experimental results across all test scenarios, including collision rate and navigation success rate analyses under varying environmental conditions.

Chapter 5 — Conclusion summarizes the key findings of this research, discusses the limitations of the proposed system, and outlines directions for future work, including potential sim-to-real transfer strategies.

This chapter has introduced the problem of low-cost autonomous robot navigation and es-

established the motivation, objectives, and significance of this research. The critical limitations of existing sensor-heavy navigation systems have been highlighted, and the case for a vision-only deep learning approach has been made. The proposed framework, grounded in ROS-Gazebo simulation and evaluated through rigorous metrics, offers a practical and scalable pathway toward affordable autonomous navigation. The subsequent chapters will build upon this foundation by reviewing existing literature, detailing the system design, and presenting the experimental outcomes.

Chapter 2

2 Background Study

This chapter presents a comprehensive review of recent peer-reviewed literature on deep learning-based vision-only autonomous robot navigation. Eight papers from Q1 and Q2 journals indexed by Scimago, published between 2023 and 2025, are critically analyzed across four dimensions: research motivation, methodology, key results, and limitations. The chapter concludes with a structured problem analysis that identifies the research gaps addressed by this project.

2.1 Literature Review

Papers were selected with a focus on monocular or RGB-only navigation, obstacle avoidance, sim-to-real transfer, and lightweight system design. Each paper is analyzed along four dimensions: research motivation, methodology, key results, and limitations.

2.1.1 MonoMPC: Monocular Vision Navigation with Learned Collision Model (IEEE RA-L, 2025)

Sharma et al. (2025) address the fundamental challenge of navigating unknown cluttered environments using only a single RGB camera. Their key insight is that monocular depth estimates from vision foundation models are too noisy for direct collision-checking — a common failure mode in prior work — and propose an alternative: using the estimated depth as contextual input to a learned probabilistic collision model. At inference, a risk-aware Model Predictive Control (MPC) planner minimizes the predicted collision risk distribution over a planning horizon.

The framework is jointly trained using both safe and unsafe trajectories, ensuring calibrated uncertainty in the collision model. Experiments demonstrate reduced collision rates and improved goal-reaching performance over several strong baselines, including

ROS Navigation Stack variants. The method is fully RGB-only and designed for deployment on lightweight ground platforms.

The primary limitation is the computational overhead of the MPC planner, which may be challenging for very low-power embedded boards. Additionally, performance under varying lighting conditions is not explicitly characterized, and the method has not been evaluated in outdoor or texture-poor environments.

2.1.2 Vision-Based Omnidirectional Navigation via Teacher-Student Monocular Depth Estimation (IEEE RA-L, 2025)

This paper proposes a privileged learning approach where a LiDAR-trained teacher policy is distilled into a student that relies solely on monocular depth maps predicted by a fine-tuned Depth Anything V2 model processing four RGB cameras. The student is trained using Proximal Policy Optimization (PPO) in simulation and deployed fully onboard an NVIDIA Jetson Orin AGX — requiring no external computation.

The student achieves navigation success rates of 82–96.5% in simulation, consistently outperforming the LiDAR-based teacher (50–89%), and demonstrates superior handling of overhanging structures and low-profile obstacles that fall outside a 2D LiDAR’s scan plane. This result is particularly significant as it demonstrates that camera-only systems can exceed LiDAR-based performance on certain obstacle types.

The primary limitation is the requirement for four cameras to achieve omnidirectional coverage, which adds cost and mechanical complexity compared to a single-camera solution. The depth estimation stage introduces inference latency, and real-world validation is confined to structured industrial settings.

2.1.3 CARE: Collision Avoidance via Repulsive Estimation for Visual Navigation (2025)

Kim et al. (2025) identify a critical vulnerability in visual navigation foundation models: while these models generalize well across environments in simulation, they fail in out-of-distribution scenes with unseen objects or different camera configurations. CARE

addresses this by adding a modular collision avoidance component that estimates repulsive forces from RGB images, steering the robot away from obstacles without requiring retraining of the base navigation policy.

The approach is validated across three distinct robot platforms — LoCoBot, TurtleBot4, and RoboMaster — demonstrating consistent collision reduction and improved collision-free travel distance in previously unseen environments. Its modularity is a key strength: it can be attached to any existing visual navigation foundation model as a plug-in safety layer.

Limitations include performance dependency on the quality of the underlying base policy and potential navigation detours in narrow corridors where repulsive forces from both walls are large. The method is evaluated exclusively in indoor environments, leaving outdoor applicability uncertain.

2.1.4 Monocular Camera-Based Complex Obstacle Avoidance via Efficient Deep RL (IEEE T-Cybernetics, 2023)

Ding et al. (2023) propose an end-to-end DRL framework for obstacle avoidance using a single monocular camera, motivated by the limitations of LiDAR-based methods in handling complex irregular obstacles such as tables, chairs, and shelves. Their custom CNN policy network processes raw RGB images and outputs velocity commands, trained entirely in simulation and tested in real environments.

The framework achieves state-of-the-art performance on complex obstacle types, demonstrating robust sim-to-real transfer and confirming that a monocular camera can capture the spatial structure needed for safe navigation in irregular scenes. The cost savings compared to laser-based systems make the approach practically viable for large-scale deployment.

Limitations include sample-inefficient training requiring extensive simulation episodes and performance degradation on highly reflective or transparent surfaces. Dynamic obstacle handling is not thoroughly evaluated, which is a significant gap for deployment in human-shared spaces.

2.1.5 Enhanced Visual SLAM for Collision-Free Driving on Lightweight Autonomous Cars (Sensors, 2024)

Lin et al. (2024) present a two-stage vision-based navigation pipeline for lightweight self-driving cars: visual SLAM for environmental perception and a modified A* algorithm for path planning with obstacle clearance constraints. The entire system operates on CPU-only hardware using a single RGB-D camera, targeting resource-constrained platforms.

The approach’s key strength is its full CPU-only operation, making it accessible to platforms without GPU acceleration. The SLAM-based map provides interpretable environmental representation, and the modified A* planner maintains safe clearance from obstacles reliably. Real hardware demonstration on a lightweight car validates practical applicability.

A notable limitation is the use of an RGB-D camera rather than a pure RGB sensor, which retains a depth hardware dependency. The A* planner is reactive rather than predictive, limiting performance with fast-moving dynamic obstacles. SLAM performance degrades in texture-poor or highly dynamic environments, which are common in real-world settings.

2.1.6 FAINT: Simulation-Trained Visual Topological Navigation Transformer (IEEE RA-L, 2025)

Suomela et al. (2025) address the question of whether simulation-trained navigation policies can match real-data-trained ones. They introduce FAINT (Fast Appearance-Invariant Navigation Transformer), a lightweight Transformer-based policy trained exclusively on simulated data with an appearance-invariant image encoder pre-trained on diverse synthetic images.

Evaluated over more than 50 hours of real-world indoor experiments on a wheeled robot and a drone, the simulation-trained FAINT outperforms its real-data-trained counterpart by 31 percentage points and surpasses the prior state of the art by 50 points in navigation success rate. This result provides strong validation for the simulation-first training paradigm adopted in the present project.

Limitations include the requirement for a pre-built topological waypoint graph, which constrains the method to partially known environments. Performance under extreme lighting variation or dense dynamic obstacles is not characterized, and Transformer inference speed on very low-end hardware is not reported.

2.1.7 IRDC-Net: Lightweight Semantic Segmentation for Monocular Robot Navigation (Sensors, 2023)

Dang et al. (2023) propose a lightweight semantic segmentation network for real-time monocular robot navigation in indoor corridor environments. Their architecture combines an FCN decoder with a quantized MobileNetV2 encoder and multi-scale feature fusion, using a Balanced Cross-Entropy loss function to handle class-imbalanced corridor datasets.

The model achieves real-time inference on a single monocular camera with significantly reduced computation compared to heavier segmentation networks, while maintaining high precision. The quantization-based compression makes it well-suited for embedded deployment. The approach is practically validated in indoor corridor navigation tasks.

Key limitations include restricted evaluation to corridor and structured indoor scenes, limiting generalizability to unstructured environments. The segmentation-based approach requires labelled training data, adding annotation overhead. Dynamic obstacles and varying lighting conditions are not systematically addressed.

2.1.8 GPS-Denied IBVS Navigation Using Low-Cost RGB Camera (IEEE RA-L, 2025)

Wang et al. (2025) propose an image-based visual servoing (IBVS) framework for UAV navigation using only an RGB camera in GPS-denied environments. Collision avoidance is achieved through AI-based monocular depth estimation (MiDaS), and multi-waypoint navigation is managed via a switching law between visual targets. The entire perception and control pipeline runs onboard a Jetson platform with no external computation.

The system’s core strength is its fully self-contained operation without GPS, LiDAR, stereo cameras, or external workstations, demonstrating the feasibility of RGB-only autonomous navigation in constrained deployment scenarios. IBVS provides precise visual target tracking without requiring an explicit map or path planner.

Limitations include the UAV-specific design, which reduces direct transferability to ground robots. MiDaS depth estimation introduces latency constraints. The framework requires clearly visible visual targets and degrades in textureless environments. Ground-level obstacle diversity is not evaluated.

Table 1 provides a comparative summary of the eight reviewed papers across key dimensions relevant to this project.

Table 1: Comparative Summary of Reviewed Papers

Paper	Sensor Input	Method	Key Result	Main Limitation
MonoMPC (2025)	RGB only	Learned collision model + MPC	Lower collision rate vs baselines	MPC overhead; no lighting test
Teacher-Student (2025)	4x RGB cameras	PPO + knowledge distillation	82–96.5% success rate	Requires 4 cameras
CARE (2025)	RGB only	Repulsive force estimation	Reduced collisions on 3 platforms	Depends on base model quality
Ding et al. (2023)	Monocular RGB	End-to-end DRL (CNN policy)	SOTA on irregular obstacles	Sample inefficient; no dynamic test
Lin et al. (2024)	RGB-D	SLAM + modified A*	CPU-only real-time operation	RGB-D needed; no dynamic obstacle
FAINT (2025)	Monocular RGB	Transformer + sim training	+50 pts over SOTA success rate	Needs topological graph
IRDC-Net (2023)	Monocular RGB	Lightweight CNN segmentation	Real-time on embedded device	Corridor-only; no dynamic/lighting
Wang et al. (2025)	RGB only	IBVS + monocular depth (MiDaS)	No GPS/LiDAR needed; onboard	UAV-specific; needs visual targets

2.2 Problem Analysis

The literature review reveals a consistent and actionable pattern of research gaps that directly motivate the design choices of the present project. The analysis is organized around four dimensions: sensor cost, environmental robustness, evaluation rigor, and simulation framework alignment.

Sensor dependency: Of the eight papers reviewed, only five (MonoMPC, CARE, Ding et al., FAINT, and Wang et al.) use a single RGB or monocular camera as the sole sensor input. The remaining papers retain a depth sensing dependency — either through multi-camera setups or RGB-D hardware. This confirms that pure RGB-only navigation at acceptable performance levels remains an active, unsolved challenge worthy of further investigation.

Lighting and environmental robustness: None of the eight reviewed papers conducts a systematic evaluation of navigation performance under varied lighting conditions — a critical real-world variable. Similarly, dynamic obstacle handling is only partially addressed by a minority of papers. This represents the most significant gap between existing literature and real-world deployment requirements, and is a primary evaluation focus of the present project.

Evaluation standardization: The reviewed papers employ inconsistent performance metrics and experimental protocols, making cross-paper comparison difficult. Only a subset explicitly report both collision rate and success rate, and none combine both metrics with a ROS-Gazebo simulation framework for controlled, reproducible evaluation. This project addresses this gap by adopting both metrics within a standardized simulation protocol.

Framework alignment: The ROS-Gazebo ecosystem is used in several reviewed works for training but is rarely used as the primary evaluation framework with systematic condition variation. This project leverages the full ROS-Gazebo stack — including sensor plugins, robot models, and physics simulation — as both the training and evaluation environment, ensuring close alignment between simulation conditions and potential real-world deployment.

In summary, the research gap addressed by this project is the absence of a lightweight, RGB-only, deep learning navigation framework that is: (i) evaluated under varied lighting and dynamic obstacle conditions; (ii) benchmarked using both collision rate and success rate; and (iii) implemented within a reproducible ROS-Gazebo simulation pipeline. The proposed framework described in Chapter 3 directly addresses each of these gaps.

Chapter 3

3 Research Methodology

This chapter describes the research methodology adopted to achieve the objectives stated in Chapter 1. It details the proposed system framework, the simulation environment configuration, the dataset and training data strategy, and the evaluation protocol.

3.1 Proposed Framework

The proposed system is a deep learning-based vision-only autonomous navigation framework for a differential-drive mobile robot. Figure 3.2 illustrates the overall architecture, which comprises four sequential modules: image acquisition, visual feature extraction, navigation policy inference, and velocity command output.

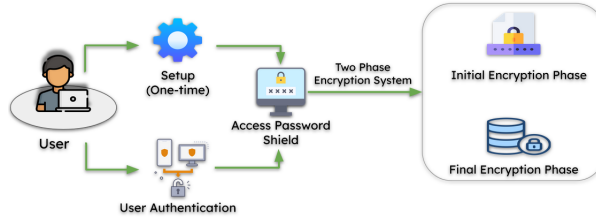


Figure 3.2: Overall System Architecture of the Proposed Navigation Framework.

3.1.1 System Architecture Overview

The framework follows an end-to-end design philosophy in which raw RGB images captured by the robot’s onboard camera are directly mapped to velocity control commands (linear and angular velocity) without intermediate 3D reconstruction, explicit mapping, or rule-based planning. The entire computation runs within the ROS computation graph, with each module implemented as a ROS node communicating via standard ROS topics and services.

The pipeline operates as follows: at each control step (target: 10 Hz), the robot’s monocular RGB camera publishes a frame to the `/camera/rgb/image_raw` topic. The

feature extraction node subscribes to this topic, processes the image through a lightweight CNN backbone, and publishes a compact feature vector to the `/nav_features` topic. The policy inference node subscribes to this feature vector and the robot’s current goal relative position (obtained from the ROS navigation stack’s goal topic), and outputs a `Twist` message to the `/cmd_vel` topic, which the robot’s ROS driver converts to wheel velocity commands.

3.1.2 CNN Backbone for Visual Feature Extraction

The visual feature extraction module employs a lightweight CNN backbone adapted from MobileNetV2, which provides an effective trade-off between feature richness and computational efficiency. MobileNetV2 uses depthwise separable convolutions that reduce the number of parameters and multiply-accumulate operations by a factor of 8–9 compared to standard convolutions of equivalent capacity, making it deployable on embedded platforms without GPU acceleration.

The backbone is initialized with weights pre-trained on ImageNet and fine-tuned during navigation policy training. Input images are resized to 224×224 pixels and normalized to the range $[-1, 1]$ prior to processing. The backbone outputs a 1280-dimensional feature vector from its final global average pooling layer, which is passed to the policy head.

3.1.3 Navigation Policy

Two candidate policy approaches are evaluated in this project, enabling comparison of supervised and reinforcement learning paradigms.

Approach A — Supervised Imitation Learning: A regression head (two fully connected layers of 256 and 64 units with ReLU activations) is appended to the CNN backbone to predict continuous linear and angular velocity commands. Training data is collected via expert teleoperation in the Gazebo simulation environment. This approach offers stable training but is limited by the quality and coverage of the collected expert demonstrations.

Approach B — Deep Reinforcement Learning (DRL) with PPO: The CNN backbone is combined with a policy network trained using Proximal Policy Optimization (PPO), a state-of-the-art on-policy DRL algorithm known for its training stability. The agent interacts with the Gazebo simulation environment through the ROS-Gazebo bridge, receiving observations (RGB images + goal position) and rewards at each step.

The reward function $R(s, a)$ is defined as:

$$R(s, a) = +r_{\text{goal}} - r_{\text{collision}} + r_{\text{approach}} \cdot \Delta d - r_{\text{rotation}} \cdot |\omega| \quad (1)$$

where $r_{\text{goal}} = 100$ (reward for reaching the goal), $r_{\text{collision}} = -50$ (penalty for collision), $r_{\text{approach}} = 1.0$ (reward proportional to reduction in distance to goal), Δd is the change in distance to goal, $r_{\text{rotation}} = 0.01$ (penalty for large angular velocity), and ω is the angular velocity command. This reward structure encourages efficient goal-directed navigation while discouraging collisions and unnecessary rotation.

3.1.4 Goal Conditioning

The navigation policy is conditioned on the robot’s relative goal position, represented as a two-dimensional vector [distance_to_goal, bearing_angle_to_goal] in the robot’s local frame. This information is provided by a lightweight goal-tracking node that subscribes to the robot’s odometry topic and the externally specified goal pose, computing the relative goal vector at each control step. Conditioning on a relative goal vector — rather than an absolute map coordinate — makes the policy invariant to the robot’s global position and eliminates the need for a global localization system.

3.2 Dataset Analysis

Given the simulation-based evaluation framework, the dataset for this project comprises two components: (i) the simulated environment configurations used for training and evaluation, and (ii) the expert demonstration data collected for the supervised imitation learning baseline.

3.2.1 Simulation Environment Configurations

Three distinct Gazebo simulation environments are constructed to support training and evaluation across varied conditions.

Environment 1 — Standard Indoor Environment: A rectangular indoor space ($10\text{m} \times 10\text{m}$) populated with 10–15 static obstacles of varied shapes (boxes, cylinders, L-shaped objects representing furniture). Lighting is set to a standard indoor illumination level (500 lux equivalent in Gazebo’s light plugin). This environment serves as the baseline training and evaluation setting.

Environment 2 — Degraded Lighting Environment: Identical layout to Environment 1 but with the ambient lighting intensity reduced to 20% of baseline (approximately 100 lux), simulating dim indoor conditions. Directional point lights are added at random positions to create uneven illumination patches, simulating flickering or partial lighting failures.

Environment 3 — Dynamic Obstacle Environment: Identical layout to Environment 1 with the addition of three dynamic obstacle agents (simulated as Gazebo model plugins) that follow randomized waypoint trajectories at walking speed (1.0–1.5 m/s). These agents simulate human pedestrians in a shared indoor space.

3.2.2 Training Data Collection

For Approach A (Imitation Learning), expert demonstration data is collected in Environment 1 using a teleoperation node that allows a human operator to drive the robot to randomly sampled goals. Each demonstration episode is recorded as a sequence of (RGB image, velocity command) pairs at 10 Hz. A total of 5,000 demonstration episodes are targeted, resulting in approximately 50,000 training samples after filtering for episodes with successful goal completion.

Data augmentation is applied to improve lighting robustness: each training image is independently augmented with random brightness adjustment (factor 0.5–1.5), random contrast adjustment (factor 0.8–1.2), and random Gaussian noise ($\sigma = 0\text{--}0.02$). These augmentations simulate the photometric variation encountered in Environments 2 and 3

and are critical for preventing overfitting to the baseline lighting conditions.

For Approach B (DRL with PPO), no expert data is required. The PPO agent is trained through autonomous interaction with the simulation environment, starting from Environment 1 and progressively increasing difficulty by introducing obstacles from Environments 2 and 3 after the agent achieves a baseline success rate of 70% in the simpler setting.

3.2.3 Evaluation Protocol

The evaluation protocol is designed to provide statistically valid performance estimates across the three test conditions. For each environment and each policy approach, 100 independent evaluation episodes are conducted. Each episode begins with the robot placed at a randomly sampled start position and given a randomly sampled goal position, with a minimum separation distance of 3 metres to ensure non-trivial navigation tasks.

Two primary performance metrics are computed:

Collision Rate (CR): The proportion of evaluation episodes in which at least one collision event is detected (defined as the robot’s laser scan or contact sensor reporting an obstacle within 0.15m).

$$CR = \frac{\text{Number of collision episodes}}{\text{Total episodes}} \quad (2)$$

Navigation Success Rate (NSR): The proportion of evaluation episodes in which the robot reaches within 0.5m of the goal position within a maximum episode duration of 60 seconds without collision.

$$NSR = \frac{\text{Number of successful episodes}}{\text{Total episodes}} \quad (3)$$

Episodes that neither collide nor reach the goal within the time limit are classified as ‘timeout’ and contribute to neither CR nor NSR numerators, reflecting real navigation scenarios where the robot becomes stuck or lost. Statistical significance of differences between approaches and conditions is assessed using a two-proportion z-test at the 0.05 significance level, with 100 episodes per condition providing sufficient power to detect

differences of 15 percentage points or greater with 80% statistical power.

3.2.4 Software and Hardware Configuration

The simulation and training pipeline is implemented using the following software and hardware stack:

- **Operating System:** ROS Noetic on Ubuntu 20.04 LTS
- **Simulator:** Gazebo 11 for physics simulation and sensor emulation
- **Deep Learning Framework:** Python 3.8 with PyTorch 2.0
- **RL Library:** Stable-Baselines3 for PPO implementation
- **ROS Packages:** `robot_state_publisher`, `rviz`, and `move_base` for robot configuration and visualization
- **Training Hardware:** NVIDIA RTX 3070 GPU (8GB VRAM) with 32GB RAM
- **Inference Validation:** CPU-only environment (2 cores at 2.0 GHz) to validate real-time performance on embedded hardware

This chapter has presented the complete research methodology adopted in this project. The proposed end-to-end vision-only navigation framework — built on a MobileNetV2 backbone and trained using either imitation learning or PPO-based deep reinforcement learning — is designed for efficient operation within the ROS-Gazebo simulation environment. Three distinct simulation environments have been defined to test standard, degraded lighting, and dynamic obstacle conditions, and a rigorous evaluation protocol using collision rate and navigation success rate has been established. The experimental results obtained using this methodology are presented and analyzed in Chapter 4.

Chapter 4

4 Implementation and Testing

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4.1 System Setup

Write System Setup in chapter4.tex

4.2 System Modules

Write System Modules all steps here in chapter4.tex

4.3 Prototype Design

Write on Prototype Design here in chapter4.tex

4.4 System Requirements for End-User

Write System Requirements for End-User in chapter4.tex

4.5 Testing & Quality Assurance

Write Testing & Quality Assurance in chapter4.tex

Write a Summary passage on this chapter here chapter4.tex

Chapter 5

5 Standards, Constraints, and Milestones

Write an Introductory passage of Standards, Constraints, and Milestones in starting of chapter5.tex

5.1 Sustainability Standards

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5.2 Impacts on Society

Write Impacts on Society in chapter5.tex

5.3 Ethics

Write Ethics in chapter5.tex

5.4 Challenges

Write Challenges in chapter5.tex

5.5 Constraints

Write different Constraints like Design constraints, Component Constraints, and Budget constraints in chapter5.tex

5.6 Timeline and Gantt Chart

Write Timeline and Gantt Chart in chapter5.tex

Write a Summary passage of Standards, Constraints, and Milestones in end of chapter5.tex

Chapter 6

6 Conclusion

Write the conclusion passage in chapter6.tex

6.1 Limitations

Write about Limitations of your work chapter6.tex

6.2 Future Works and Direction

Write Future Works and Direction in chapter6.tex

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